

Jurabek Murodov

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EDUCATION

Stony Brook University

Expected Graduation May 2027

Bachelor of Engineering, Computer Engineering

GPA: 3.82

Relevant Coursework: Data structures, Software Techniques, Discrete Mathematics, Digital Logic Design, Digital Design Using VHDL and PLDs, Programming in C, Embedded Microcontroller Systems, Electrical Circuit Analysis

SKILLS

- Programming: Java, C, C++, C#, HTML, CSS, JavaScript, Python, and Arduino
- Software: Visual Studio, NetBeans, Microsoft Excel, OrCad, Autodesk Fusion, OnShape, KiCad, and LTspice
- Languages: English, Tajik, Persian, Russian

PROJECTS

Intelligent Pathfinding and Memory Simulation in C

- Constructed a memory-efficient stack-based navigation system, enabling a virtual agent to autonomously traverse complex mazes with optimized decision-making.
- Formulated an algorithmic backtracking method leveraging Abstract Data Types (ADT) to enhance computational efficiency and adaptive problem-solving.

Omni-Directional Car

- Designed and integrated a 3-wheeled omnidirectional drive system, leveraging ESP32 microcontroller and PWM motor drivers to execute seamless lateral, diagonal, and rotational movements.
- Programmed a real-time Bluetooth control interface via the Dabble app, optimizing response latency and user-command execution through adaptive signal processing.

Arduino-Based Car

- Assembled a 2-wheeled autonomous vehicle, utilizing OnShape for CAD modeling and 3D printing techniques to fabricate a lightweight, structurally optimized chassis.
- Calibrated an adaptive navigation system, integrating light-detection sensors with real-time path correction algorithms to enhance obstacle avoidance and environmental responsiveness.

Marble Sorting System

- Implemented a precision sorting device with an Arduino microcontroller, incorporating two micro servos and an infrared sensor for the segregation of colored from clear marbles.
- Optimized control logic for instant sorting, showcasing advanced programming and sensor integration.

Smart USB Charger

- Developed a multi-functional USB charging station with an integrated real-time clock, utilizing a four-digit seven-segment LED display to ensure precise and continuous timekeeping during power cycles.
- Engineered a binary-encoded LED feedback system, soldering a user-friendly interface with six LEDs to visually represent elapsed seconds, enhancing accessibility and real-time monitoring of time progression.

Personal Website

- Utilized HTML, CSS, and JavaScript to build a responsive and visually appealing portfolio, incorporating interactive elements and streamlined code for optimal performance.

EXPERIENCE

IEEE Stony Brook Robotics Micromouse

December 2024 - Present

Embedded Systems Lead

Stony Brook, NY

- Lead the development of embedded systems for the SBU Micromouse Team, a competition-driven project focused on designing an autonomous robotic mouse capable of efficiently navigating a maze.
- Collaborate with teammates to refine hardware-software interactions, optimizing navigation and control precision through real-time sensor feedback.
- Implement time-of-flight sensors and encoders, utilizing the VL53L0X API for accurate distance measurement and localization, improving maze-solving performance.

Stony Brook University Campus Residences

August 2024 - Present

Resident Assistant

Stony Brook, NY

- Oversee college residence hall floor of diverse undergraduate students ranging from international to transfer students, by developing a community and ensuring each individual is comfortable in their living environment.
- Plan and implement programs for residents and maintain all bulletin boards and fliers with accurate, up to date, and useful information.
- Mediate roommate and suitemate conflict with the use of problem-solving and communication skills.